

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) who uses Claude Code daily as an AI coding assistant to design and accelerate automation solutions, most recently at Deutsche Bundesbank. Comfortable researching a problem, prototyping a solution with AI-assisted tooling, and iterating based on feedback, with a foundation in Python scripting, prompt-driven workflows, and GitHub-based collaboration.

Core Competencies

- **AI-Assisted Development:** Daily use of Claude Code for prototyping, automation, and workflow design; prompt engineering to translate requirements into working solutions
- **Automation Solutions:** Built configuration-driven automation replacing manual, case-by-case processes at Deutsche Bundesbank
- **Programming:** Python, SQL, R, Java, C++; Git/GitHub version control
- **Tooling:** MS Office (Excel, PowerPoint, SharePoint), Microsoft Copilot, Claude Code, GitHub
- **Collaboration:** Agile Scrum experience (sprint planning, daily stand-ups, sprint reviews); cross-functional stakeholder communication
- **Languages:** English (C1, TOEFL iBT 105/120), German (C1), Spanish (native)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations), with automated Jira ticketing and emails to reporting banks
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Automated Validation Framework for Regulatory Reporting Data – converted a non-scalable, case-by-case validation script into a parameterized, reusable Python solution

 - Designed parameterized queries applicable to any database, attribute, and reporting period, replacing hard-coded per-case script logic
 - Built a YAML-driven configuration interface enabling batch validation across multiple databases, attributes, and reporting periods in a single call

Test Data Generator for the Bundesbank Platform – configuration-driven system producing complete, consistent synthetic test datasets

 - Generated Cartesian-product attribute combinations per table with consistent IDs across tables to preserve referential integrity

Projects

- **Personal Project (GitHub, built with Claude Code)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts), using Claude Code throughout to accelerate implementation of the SQLite schema, cleaning logic, and trigger-based automation while retaining full ownership of design decisions
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; 5 quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication

Education

- **Tecnológico de Monterrey** Mexico
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, numerical optimization.
- **Universität Göttingen** Germany
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.